Hydration Effects on Temperature Regulation

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During exercise in the heat, sweat output often exceeds water intake which results in a body water deficit (hypohydration) and electrolyte losses. Daily water losses can be substantial and persons need to emphasize drinking during exercise as well as at mealtime. Aerobic exercise tasks are likely to be adversely affected by heat stress and hypohydration; and the warmer the climate the greater the potential for performance decrements. Hypohydration increases heat storage and reduces one's ability to tolerate heat strain. The increased heat storage is mediated by reduced sweating rate (evaporative heat loss) and reduced skin blood flow (dry heat loss) for a given core temperature. Hyperhydration (increased total body water) has been suggested to reduce physiologic strain during exercise – heat stress, however, data supporting that notion are not robust.

Key words: Core temperature, electrolyte requirements, skin blood flow, dehydration, fluid redistribution, hypohydration, hyperhydration, sweating, thermoregulation, water requirements.

Introduction

Depending on the climatic conditions the relative contributions of evaporative and dry (radiative and conductive) heat exchange to the total heat loss will vary (8). The hotter the climate the greater the dependence on evaporative heat loss, and thus on sweating. Therefore, a substantial volume of body water may be lost via sweating to enable evaporative cooling in hot climates. Generally, the individual dehydrates during exercise because of fluid nonavailability or a mismatch between thirst and body water requirements (2,3). In these instances, the individual starts the exercise task as euhydrated but incurs an exercise-heat mediated dehydration over a prolonged period of time.

Fluid and Electrolyte Needs

A person's sweating rate is dependent upon the climatic conditions, clothing worn and exercise intensity (1,8). Persons in desert climates often have sweating rates of 0.3 to 1.2 $L \cdot h^{-1}$ while performing occupational activities (1). Persons wearing protective clothing often have sweating rates of 1 to 2 $L \cdot h^{-1}$ while performing light intensity exercise. Likewise, athletes performing high intensity exercise commonly have sweating rates of 1.0 to 2.5 $L \cdot h^{-1}$ while in the heat. Fluid requirements will vary in relation to climatic heat stress, clothing worn, acclimation state and physical activity levels. Daily fluid requirements might range (for sedentary to very active persons) from 2–4 L/day in temperate climates and from 4–10 L/day in hot climates (2,3). Electrolytes, primarily sodium chloride and to a lesser extent potassium, are lost in sweat. Sweat sodium concentration averages ~40 mEq/L (range 10–100 mEq/L) and varies depending upon diet, sweating rate, hydration and heat acclimation level. Heat acclimated persons have relatively low sodium concentrations (>50% reduction) in sweat (8).

During exercise-heat stress, a principle problem is to avoid dehydration by matching fluid consumption to sweat loss (4). This is a difficult problem because thirst does not provide a good index of body water requirements. Thirst is probably not perceived until an individual has incurred a water deficit of ~2% of body weight. Numerous investigators report that *ad libitum* water intake results in incomplete water replacement or "voluntary" dehydration during exercise and/or heat exposure. The flavoring and cooling of ingested fluid increases its palatability and can help to minimize "voluntary" dehydration (3). Heat acclimation status may also influence the "voluntary" dehydration incurred during exercise in the heat. Although heat acclimation improves the relationship between thirst and body water needs "voluntary" dehydration still occurs. Since thirst provides a poor index of body water needs, persons will dehydrate by 2% to 8% of their body weight during situations of prolonged sweat loss.

Exercise Performance

Numerous studies have examined the influence of hypohydration less than normal/total body water on maximal aerobic power and physical work capacity (5,6). In temperate climates, a body water deficit of less than 3% BWL does not alter maximal aerobic power. Maximal aerobic power has been reported as being decreased when hypohydration equaled or exceeded

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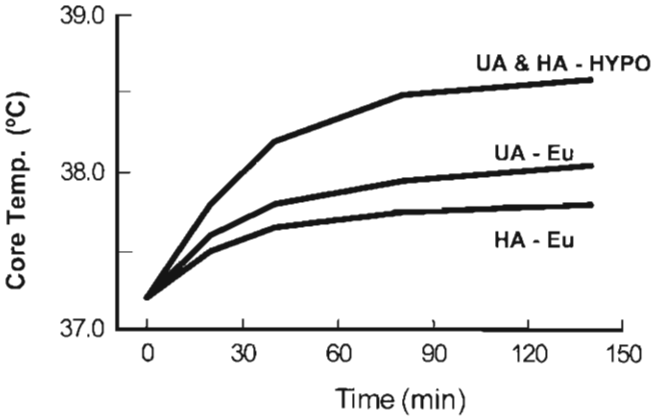

Fig. 1 Core temperature responses during exercise – heat stress in euhydrated (Eu) and hypohydrated (HYPO) (5% BWL) persons both before (UA) and after (HA) being heat acclimated (7).

3% BWL. Therefore, a critical water deficit (3% BWL) might exist before hypohydration reduces maximal aerobic power in temperate climates. In hot climates, small (2% BWL) to moderate (4% BWL) water deficits result in large reduction of maximal aerobic power. Physical work capacity decreased by marginal (1%–2% BWL) water deficits and the reduction is larger with increasing water deficit. Clearly, hypohydration resulted in much larger decrements of physical work capacity in hot as compared to temperate climates. It appears that the thermoregulatory system, perhaps via increased body temperatures, has an important role in the reduced aerobic exercise performance mediated by a body water deficit (5, 6).

Hypohydration decreases tolerance time for exercise – heat stress as well as physiologic tolerance. Sawka and colleagues (9) had subjects attempt treadmill walks (~25% $\dot{V}O_2$max for 140 min) in a hot-dry (T_a = 49 °C, rh = 20%) environment when euhydrated and when hypohydrated by 3%, 5%, and 7% BWL. All eight subjects completed the euhydration and 3% hypohydration experiments, and seven subjects completed the 5% hypohydration experiments. For the 7% hypohydration experiments, six subjects discontinued after completing only (mean) 64 minutes. Clearly, hypohydration increases the incidence of exhaustion from heat strain.

To address whether hypohydration alters physiologic tolerance to heat strain, Sawka and colleagues (10) had subjects walk to voluntary exhaustion when either euhydrated or hypohydrated (8% of total body water). The experiments were designed so that the combined environment (T_a = 49 °C, rh = 20%) and exercise intensity (47% $\dot{V}O_2$max) would not allow thermal equilibrium and heat exhaustion would eventually occur. Hypohydration reduced tolerance time (121 to 55 min), but more importantly, hypohydration reduced the core temperature that a person could tolerate. Heat exhaustion occurred at a core temperature ~0.4 °C lower when hypohydrated than when euhydrated. These findings suggest that hypohydration not only impairs exercise performance, but also reduces physiologic tolerance to heat strain.

Hypohydration and Temperature Regulation

Hypohydration increases core temperature responses during exercise in temperate and hot climates (5, 6). A critical deficit of 1% of body weight elevates core temperature during exercise. As the magnitude of water deficit increases, there is a concomitant graded elevation of core temperature during exercise heat stress. The magnitude of core temperature elevation ranges from 0.10 to 0.23 °C for every percent body weight lost, and this elevation is greater during exercise in hot than temperate climates. Recent studies at our laboratory indicate that the core temperature elevation is not affected by exercise intensity.

Hypohydration not only elevates core temperature responses, but it negates the core temperature advantages conferred by high aerobic fitness and heat acclimation (7). Fig. 1 illustrates the effects of hypohydration (5% BWL) on core temperature responses in the same persons when unacclimated and when heat acclimated (7). Heat acclimation lowered core temperature responses when euhydrated; however, when hypohydrated similar core temperature responses were observed regardless of acclimation state. Therefore, the core temperature penalty induced by hypohydration was greater in heat acclimated than unacclimated persons. These data indicate that heat acclimated persons (who have increased sweating rates) who do not drink adequately may more rapidly experience the adverse effects of hypohydration than their nonacclimated counterparts.

Hypohydration impairs both dry and evaporative heat loss (or, if the air is warmer than the skin, dehydration aggravates dry heat gain). Hypohydration delays sweating onset and skin vasodilatation. It also reduces sweating sensitivity. Hypohydration may be associated with either reduced or unchanged sweating rates at a given metabolic rate in the heat. The physiological mechanisms mediating the reduced dry and evaporative heat loss from hypohydration include both the separate and combined effects of plasma hyperosmolality and reduced blood volume (5).

Hyperhydration

Hyperhydration, or greater than normal body water, has been suggested to improve, above euhydration levels, thermoregulation and exercise-heat performance (6). The concept that hyperhydration might be beneficial for exercise performance arose from the adverse consequences of hypohydration. It was theorized that body water expansion might reduce the cardiovascular and thermal strain of exercise by expanding blood volume and reducing blood tonicity; thereby improving exercise performance. Studies that have directly expanded blood volume (e.g. infusion) have usually reported decreased cardiovascular strain during exercise, but have disparate results on heat dissipation and exercise-heat performance (5, 6). Studies that have attenuated plasma hyperosmolality during exercise-heat stress generally report improved heat dissipation, but have not addressed exercise performance (5, 6).

Numerous studies have evaluated hyperhydration effects on thermoregulation in the heat. Some studies observe smaller core temperature increases during exercise with hyperhydration, however, most of those studies suffer from confounding

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experimental procedures (6). Examination of their data indicates that "control" conditions generally do not represent euhydration, and that there may have been order problems so that subjects were more heat acclimated during hyperhydration trials (6). Recent studies at our laboratory have controlled for these confounding factors, and observed no thermal advantage with either water hyperhydration or glycerol hyperhydration during exercise-heat stress.

Key Points

- Heat stress and physical exercise elicits sweating and both increase daily fluid requirements.
- Thirst sensations underestimate body water needs so persons incur body water deficits during exercise – heat stress.
- Hypohydration reduces aerobic exercise performance and decrements are greater in hot than temperate climates.
- Hypohydration increases heat strain because heat loss (by both sweating and skin blood flow) responses are adversely affected.
- Hyperhydration probably has marginal affect on heat strain and exercise – heat performance.

Disclaimer

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